



AJITH SAMPATH

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🐙 Github ID ORCID in LinkedIn 🔗 WebPage

CORE EXPERTISE

- **Scientific programming** in Python for large-scale data analysis
- **Radio astronomy** and 21cm cosmology (calibration and signal extraction)
- **Machine learning** for scientific data (classification, signal detection, foreground removal).
- Development of **scalable pipelines and algorithms** for large datasets (HIRAX analysis pipeline)

TECHNICAL SKILLS

- **Programming:** Python (NumPy, SciPy, Pandas, Matplotlib)
- **Machine Learning:** Scikit-learn, PyTorch; classification, signal detection, feature extraction, foreground removal (21cm cosmology)
- **Data Processing:** Large-scale data analysis, scalable pipeline development, data cleaning, and transformation
- **Scientific Computing:** Numerical methods, statistical analysis, signal processing
- **High-Performance Computing:** Parallel computing (MPI), HPC clusters, job scheduling (SLURM)
- **Radio Astronomy Tools:** CASA (calibration, imaging, simulation), WSCLEAN (wide-field imaging), AOFlagger (RFI detection and mitigation), CST Studio Suite (electromagnetic RF simulations)
- **Tools:** Git, Linux, LaTeX

PROFESSIONAL EXPERIENCE

POSTDOCTORAL FELLOW

Jun 2025 – Present

Department of Theoretical Physics, University of Geneva, Switzerland

- SNSF Lead Agency grant (No. 10001432)
- Developed an empirical primary beam model for high-precision HIRAX calibration
- Contributed to large-scale simulations and data analysis pipelines in HIRAX
- Applied machine learning for foreground modeling and mitigation in 21cm cosmology

INTERN

Jan 2020 – Jun 2020

IIT Indore, India

- Performed imaging-based 21cm power spectrum estimation using MWA data

VISITING STUDENT

Jul 2019 – Dec 2019

Indian Institute of Astrophysics, India

- Developed a CASA-based imaging pipeline for the GRAPH telescope
- Built a GUI analysis tool for the GLOSS spectrometer

EDUCATION

UNIVERSITY OF KWAZULU-NATAL, SOUTH AFRICA | *PhD* 2021 – 2024

- Thesis title: Beam Calibration and Foreground Leakage in Intensity Mapping experiments.
- Supervisor: Prof. Kavilan Moodley
- NRF - TWAS Doctoral Fellowship

UNIVERSITY OF MADRAS, CHENNAI, INDIA | | *MSc Theoretical Physics* 2017 – 2019

- Related Coursework: Astrophysics, Theoretical Cosmology, Numerical Methods.
- Final semester short Project: HI observations of Barred Spiral Galaxies.

BHARATHIDASAN UNIVERSITY, TAMIL NADU, INDIA | | *BSc Physics* 2014 – 2017

- Gold Medalist in Physics - A.V.C. College (Autonomous).

AWARDS

- **"Biggest Impact"** award for Oral Presentation - with cash prize ZAR 5000 2023
Postgraduate Research and Innovation Symposium 2023,
University of KwaZulu-Natal, South Africa.
- **"Winner"** of BRICS Astronomy Hackathon on Machine Learning 2023
BRICS Astronomy meeting, Capetown, South Africa.

LANGUAGE PROFICIENCY

- **English** (Academic IELTS 7.5 - **C1**), French (A1-Learning) and Tamil (Native language).
- Had a **German** introductory course in my Master's.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

IMPRS-HD "AI for Astronomy" Summer School, Heidelberg, Germany.	Sep 2025
BRICS Astronomy Hackathon, IDIA, Cape Town, South Africa.	Oct 2023
CASA-VLBI Workshop - JIVE, Virtual.	Nov 2020
Synthesis Imaging Workshop - NRAO, Virtual.	Jul 2020
Observing First Billion years of the Universe School, IIT Indore, India	Jan 2020
Solar Astronomy Winter School, Kodaikanal Radio Observatory, India.	Jan 2019
"Astrophysics: An observational view of the Universe" Summer Fellowship, Indian Academy of Sciences, Bengaluru, India.	May-Jun 2018

PUBLICATIONS

1. Primary Beam Chromaticity in HIRAX: I. Characterization from Simulations and Power Spectrum Implications 2024
Ajith Sampath, Devin Crichton, Kavilan Moodley, et al.
The Astrophysical Journal - DOI: 10.3847/1538-4357/ae1b89
2. The HIRAX: Overview and Status Update. 2024
Anthony Walters, ... **Ajith Sampath** ... et al. 18th European Conference on Antennas and Propagation
(EuCAP), Glasgow, UK, 2024, pp. 1-5, DOI:10.23919/EuCAP60739.2024.10501639

3. HIRAX: 256-element array status and overview. 2022
*Devin Crichton, ... **Ajith Sampath** ... et al. Journal of Astronomical Telescopes, Instruments, and Systems*
 Vol. 8, Issue 1, 011019 <https://doi.org/10.1117/1.JATIS.8.1.011019>
4. Antenna characterization for the HIRAX experiment. 2022
*Emily R.Kuhn, ... **Ajith Sampath** ... et al. Ground-based and Airborne Telescopes IX. Vol. 12182.*
 SPIE, 2022. <https://doi.org/10.1117/12.2627856>

KEY PROJECTS

- **HIRAX Primary Beam characterization:** Analytically modeled, EM-simulated, and measured antenna patterns, and characterized the impact on the power spectrum space.
- **HIRAX Data Analysis Pipeline:** Contributed to the development of scalable data analysis pipelines for large radio astronomy datasets. Implemented algorithms for signal extraction and foreground mitigation in 21cm cosmology
- **Machine Learning for Foreground Mitigation:** Developed and applied ML models for classification and removal of foreground contamination. Improved accuracy of cosmological signal recovery from noisy data.
- **GRAPH Imaging Pipeline (CASA):** Built an automated imaging pipeline for the GRAPH solar radio telescope using CASA. Processed and analyzed solar interferometric data.
- **GLOSS Spectrometer Analysis Tool:** Developed a GUI-based tool for data visualization and analysis of spectrometer observations - including solar bursts model fitting.

CONFERENCE CONTRIBUTIONS

1. The significance of characterizing the sidelobe systematics in 21cm Intensity Mapping experiments: The HIRAX case. 2024
Ajith Sampath, Devin Crichton, & Kavilan Moodley. DOI: 2024IAUGA..32P.517S
2. Full-wave analysis of mutual coupling in the HIRAX radio telescope. 2022
*Q Gueuning, D Crichton, **A Sampath**, AK Brown, E de Lera Acedo & K Moodley.*
International Conference on Electromagnetics in Advanced Applications (ICEAA), Cape Town, South Africa, 2022, pp. 288-288, DOI: 10.1109/ICEAA49419.2022.9899905.

SELECTED CONTRIBUTED TALKS

- **Swiss Cosmology Days** 2026 — Precision Cosmology meets Imperfect Instruments
- **Cosmology on Alps** 2026 — HIRAX Primary Beam: Challenges and Updates
- **SKACH Winter Meeting** 2026 — HIRAX Overview and Current Status
- **Swiss SKA Days** 2025 — Primary Beam Chromaticity in HIRAX
- **SARAO Postgraduate Conference** 2023&2024 — Beam Systematics and Primary Beam Modelling
- **UKZN PRIS Symposium** 2023 — Instrument Characterization (Biggest Impact Award)
- **Cosmology on Safari.** 2023 - An Analytical approach to characterize Beam Systematics in HIRAX

INVITED TALKS

- **Invited Talk - Alumni Lecture series** Jan 2019
Invited by the department of Physics, A.V.C. College for a lecture on "Introduction to Physics of Cosmos".

MENTORSHIP & COMMUNITY ENGAGEMENT

- **Student supervision** 2026
Supervised a Master's student on a Semester project titled "Cosmology with Fast Radio Bursts" at the University of Geneva
Student: Swan Dubief (Swan.Dubief@etu.unige.ch)
- **Organised Department Seminars** 2023-2024
Co-organizer of Astronomy Research Center Weekly seminars for three semesters at the University of KwaZulu-Natal, South Africa.
- **Medical Data Analysis** 2023-2024
Worked on ICU Heart Rate and Blood pressure data analysis on the KREATE community engagement program with eThekweni Hospital, Durban, South Africa.
- **Led the Weekly KREATE Computer Literacy workshop** 2023
For underprivileged high school students from Lindelani Youth Forum and Community Life Center, Durban, South Africa.
- **Co-led two KREATE Machine Learning workshops** 2023
For University students and industry staff hosted by SmartXchange (a business incubator and innovation hub), Durban, South Africa.

PERSONAL INTERESTS

Chess Active player on Chess.com with an interest in strategic problem-solving.

Boxing & Fitness Amateur Boxing training (Boxing Club Genevois, Geneva).

Photography Travel photography enthusiast with a focus on digital post-processing.

REFEREES

- **Prof. Martin Kunz**
University of Geneva, Geneva, Switzerland.
✉ Martin.Kunz@unige.ch
- **Prof. Kavilan Moodley**
University of KwaZulu-Natal, Durban, South Africa.
✉ Moodleyk41@ukzn.ac.za